

# ASG CAD MASTER STANDARD - AutoCAD Environment for Aluminium, Glass & Facade Engineering

Standard R1 | 10.10.2026 | Supersedes ASG-STD-CAD-001 R0 and the earlier cad-standard / asg-master-template packages. Conventions are informed by ISO 128 (line widths), ISO 3098 (lettering heights), ISO 216 (paper sizes), ISO 5457 (borders), ISO 5455 (scales) and ISO 7200 (title block data). No formal compliance or certification is claimed - the ISO texts were not available for verification.

## 1. Design philosophy

1. Model geometry is drawn full size, 1:1, in millimetres.
2. Geometry lives in model space; annotation is annotative in model space or true-size in paper space; sheets live in layouts.
3. Layers are meaningful, coded and documented; object properties are ByLayer.
4. Blocks have predictable names, datum base points and mm insertion units.
5. Dimensions always report true geometry - dimension text is never overridden.
6. Annotation is sized by its plotted paper height (ISO 3098 series), never guessed in model space.
7. Layouts use genuine ISO 216 paper sizes and locked, controlled viewports.
8. Plots are monochrome with a fixed ISO 128 lineweight hierarchy - predictable on every printer and PDF.
9. Proprietary manufacturer data is never invented.
10. Easy to learn, hard to misuse: few styles, clear names, one setup command, one QA command.

## 2. Package structure

| Folder                         | Content                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00_START_HERE.pdf              | Two-page picture guide: install in 2 minutes, then every new drawing in 6 steps                                                                                                   |
| 01_Master_Template             | ASG_Master_Template.dxf (master) + ASG_Install.lsp (one-drag installer, setup and QA). The installer saves ASG_Master_Template.dwg here and the DWT in AutoCAD's Template folder. |
| 02_Drawing_Templates           | One single-layout template per sheet size (DXF)                                                                                                                                   |
| 03_Page_Setups                 | ASG_Page_Setups.csv - page setup definitions (created as named page setups by the installer)                                                                                      |
| 04_Plotting_Standards          | ASG_Monochrome.ctb                                                                                                                                                                |
| 05_CAD_Standards_Documentation | ASG_CAD_Standards.pdf (this document), ASG_CAD_Quick_SOP.pdf (one-page drafting SOP for every draftsman) + layout previews                                                        |
| 06_Layer_Register              | ASG_Layer_Register.xlsx / .csv                                                                                                                                                    |
| 07_QA_Validation               | Validation report PDF / TXT, temporary QA drawing, QA renders                                                                                                                     |
| 08_Automation_Source           | asg_standard.py (all definitions), build_all.py, build_sop.py, build_guide.py, qa_validate.py                                                                                     |
| 09_Team_Kit                    | What every team member needs: installer, print setting, SOP, guide - the installer adds the DWT                                                                                   |

### 3. Units and drawing-resident variables written into the template

| Variable      | Value                | Purpose                                                               |
|---------------|----------------------|-----------------------------------------------------------------------|
| INSUNITS      | 4                    | Insertion units = millimetres (no unit conversion on block insert)    |
| MEASUREMENT   | 1                    | Metric - acadiso.lin / acadiso.pat                                    |
| LUNITS        | 2                    | Linear units = Decimal                                                |
| LUPREC        | 0                    | Linear display precision 0 dp                                         |
| AUNITS        | 0                    | Angular units = Decimal Degrees                                       |
| AUPREC        | 2                    | Angle display precision 0.00                                          |
| ANGBASE       | 0.0                  | 0 deg = East                                                          |
| ANGDIR        | 0                    | Counter-clockwise positive                                            |
| LIMMIN        | (0.0, 0.0)           | Nominal limits origin                                                 |
| LIMMAX        | (200000.0, 150000.0) | Nominal 200 x 150 m - never restrictive                               |
| LIMCHECK      | 0                    | Limits not enforced (large facade elevations)                         |
| LTSCALE       | 1.0                  | Global linetype scale 1 - dash lengths in paper mm                    |
| PSLTSCALE     | 1                    | Viewport scale controls paper-space linetype scaling                  |
| CELTSCALE     | 1.0                  | New-object linetype scale 1                                           |
| LWDISPLAY     | 1                    | Lineweights displayed on screen                                       |
| CELWEIGHT     | -1                   | New objects: lineweight ByLayer                                       |
| CECOLOR       | 256                  | New objects: colour ByLayer                                           |
| DIMASSOC      | 2                    | Associative dimensions follow geometry                                |
| PLINEGEN      | 1                    | Continuous linetype pattern along polylines                           |
| FILLMODE      | 1                    | Fills displayed                                                       |
| MIRRTEXT      | 0                    | Text not mirrored                                                     |
| TEXTSIZE      | 2.5                  | Default text height (paper mm for annotative styles)                  |
| PSTYLEMODE    | 1                    | Colour-dependent plot styles (CTB)                                    |
| WORLDVIEW     | 1                    | UCS follows WCS in views                                              |
| UCSORG        | (0.0, 0.0, 0.0)      | UCS origin = WCS origin                                               |
| UCSXDIR       | (1.0, 0.0, 0.0)      | UCS = World                                                           |
| UCSYDIR       | (0.0, 1.0, 0.0)      | UCS = World                                                           |
| VISRETAIN     | 1                    | Xref layer overrides retained                                         |
| PROXYGRAPHICS | 1                    | Proxy images saved (interoperability)                                 |
| CLAYER        | ASG-OUTLINE-PRIMARY  | Default current layer (plottable geometry)                            |
| TEXTSTYLE     | ASG-TEXT-ANNO        | Default text style                                                    |
| DIMSTYLE      | ASG-DIM-ANNO         | Default dimension style (header dim vars synchronised - no overrides) |
| CMLSTYLE      | ASG-MLINE-2          | Default multiline style                                               |

Drawing-resident variables set inside AutoCAD by the installer (saved with the DWG / DWT):

| Variable       | Value              | Purpose                                                                     |
|----------------|--------------------|-----------------------------------------------------------------------------|
| CANNOSCALE     | 1:1                | Current annotation scale in model space                                     |
| MSLTSCALE      | 1                  | Model-space linetypes follow the annotation scale                           |
| ANNOALLVISIBLE | 0                  | Show only annotation that supports the current scale (no visual duplicates) |
| CMLEADERSTYLE  | ASG-ML-ANNO        | Default multileader style                                                   |
| CTABLESTYLE    | ASG-TABLE-STANDARD | Default table style                                                         |
| HPLAYER        | ASG-HATCH          | New hatches go to ASG-HATCH automatically                                   |

| Variable  | Value         | Purpose                                          |
|-----------|---------------|--------------------------------------------------|
| DIMLAYER  | ASG-DIMENSION | New dimensions go to ASG-DIMENSION automatically |
| UCSFOLLOW | 0             | View does not rotate with UCS changes            |

Precision: ordinary dimensions 0 dp; fabrication details ASG-DIM-DETAIL 0.0; angles 0.00. INSUNITS = mm on the drawing and every block - no unit conversion on insertion.

#### 4. Machine / user-profile settings (NOT stored in any DWG / DWT)

These live in each PC's AutoCAD profile (registry). A template cannot enforce them. Recommended values, applied only by the opt-in command ASG-PROFILE.

| Variable         | Recommended | Purpose                                                                  |
|------------------|-------------|--------------------------------------------------------------------------|
| OSMODE           | 2223        | Endpoint, Midpoint, Centre, Node, Intersection, Perpendicular, Extension |
| AUTOSNAP         | 63          | Marker, magnet, tooltip, aperture, polar + object snap tracking          |
| POLARANG         | 45.0        | Polar tracking increment 45 deg                                          |
| ORTHOMODE        | 0           | Ortho off - polar tracking preferred                                     |
| DYNMODE          | 3           | Dynamic input: pointer + dimension input                                 |
| DYNPROMPT        | 1           | Prompts near cursor                                                      |
| SELECTIONCYCLING | 2           | Cycle overlapping objects with list                                      |
| GRIPS            | 1           | Grips on                                                                 |
| GRIPSIZE         | 5           | Grip size                                                                |
| PICKBOX          | 4           | Selection target size                                                    |
| APERTURE         | 8           | Object snap target size                                                  |
| PICKFIRST        | 1           | Noun-verb selection                                                      |
| PICKADD          | 2           | Shift removes from selection                                             |
| ANNOAUTOSCALE    | 0           | Do NOT auto-add scales to annotative objects                             |
| SAVETIME         | 10          | Autosave every 10 min                                                    |
| ISAVEBAK         | 1           | Keep .bak on save                                                        |
| LAYOUTREGENCTL   | 2           | Fast layout switching                                                    |
| REFPATHTYPE      | 1           | Relative xref paths                                                      |

## 5. Layer dictionary

Naming: ASG-<FUNCTION>[-<QUALIFIER>] in full words. Codes: G geometry, T technical, A annotation, S sheet. All objects ByLayer. Layer 0 for block definitions only. Defpoints is an AutoCAD system layer (never plots).

| Code | Layer                 | ACI | Linetype   | LW   | Plot | Recommended use                                                   |
|------|-----------------------|-----|------------|------|------|-------------------------------------------------------------------|
| G01  | ASG-OUTLINE-PRIMARY   | 7   | Continuous | 0.35 | Yes  | Overall frame / panel / opening outlines in elevation and plan    |
| G02  | ASG-OUTLINE-SECONDARY | 8   | Continuous | 0.18 | Yes  | Edges beyond, minor returns, internal lines                       |
| G03  | ASG-ALUMINIUM-PROFILE | 7   | Continuous | 0.25 | Yes  | Frames, mullions, transoms, sashes, beads (elevation and section) |
| G04  | ASG-GLASS             | 4   | Continuous | 0.18 | Yes  | Glass / mirror edges, DGU and laminate ply lines                  |
| G05  | ASG-STAINLESS-STEEL   | 7   | Continuous | 0.25 | Yes  | SS fittings, spigots, channels, handrails, patch plates           |
| G06  | ASG-MILD-STEEL        | 7   | Continuous | 0.25 | Yes  | MS brackets, sub-frames, support steel, base plates               |
| G07  | ASG-HARDWARE          | 8   | Continuous | 0.18 | Yes  | Hinges, handles, locks, rollers, closers, patch fittings          |
| G08  | ASG-FIXING            | 8   | Continuous | 0.18 | Yes  | Anchors, screws, bolts, rivets, fixing brackets                   |
| G09  | ASG-GASKET            | 8   | Continuous | 0.13 | Yes  | EPDM gaskets, brushes, setting / distance blocks                  |
| G10  | ASG-SEALANT           | 8   | Continuous | 0.13 | Yes  | Structural / weather silicone, backer rod, tapes                  |
| G11  | ASG-INSULATION        | 252 | Continuous | 0.13 | Yes  | Insulation and thermal breaks (plots grey)                        |
| G12  | ASG-CLADDING          | 7   | Continuous | 0.25 | Yes  | ACP / solid panels, flashings, copings                            |
| T01  | ASG-HIDDEN            | 8   | HIDDEN2    | 0.18 | Yes  | Concealed edges, embedded parts                                   |
| T02  | ASG-CENTERLINE        | 8   | CENTER2    | 0.13 | Yes  | Axes of symmetry, hole centres, module lines                      |
| T03  | ASG-CUTTING-PLANE     | 7   | PHANTOM2   | 0.35 | Yes  | Section cut lines (section markers go on ASG-SYMBOL)              |
| T04  | ASG-SECTION-CUT       | 7   | Continuous | 0.50 | Yes  | Outline of material cut by the section plane - heaviest geometry  |
| T05  | ASG-HATCH             | 252 | Continuous | 0.09 | Yes  | Pattern hatches in sections (plots grey)                          |
| T06  | ASG-HATCH-FILL        | 251 | Continuous | 0.09 | Yes  | Solid fill of thin aluminium / steel walls (plots dark grey)      |
| T07  | ASG-SETOUT            | 8   | DASHDOT2   | 0.13 | Yes  | Opening lines, setting-out lines that must print                  |
| T08  | ASG-REFERENCE         | 253 | Continuous | 0.13 | Yes  | Architectural / structural background, xrefs (plots light grey)   |
| T09  | ASG-CONSTRUCTION      | 6   | Continuous | 0.00 | NO   | Temporary helper lines only - never final geometry                |
| A01  | ASG-DIMENSION         | 3   | Continuous | 0.13 | Yes  | All dimensions                                                    |
| A02  | ASG-TEXT              | 7   | Continuous | 0.18 | Yes  | Labels, view titles                                               |
| A03  | ASG-TEXT-NOTE         | 7   | Continuous | 0.18 | Yes  | General and specification notes                                   |
| A04  | ASG-MULTILEADER       | 3   | Continuous | 0.13 | Yes  | Callouts and material labels                                      |
| A05  | ASG-TABLE             | 7   | Continuous | 0.18 | Yes  | Schedules, registers, note tables                                 |
| A06  | ASG-LEVEL             | 3   | Continuous | 0.13 | Yes  | FFL / SSL / sill / head level markers                             |
| A07  | ASG-GRID              | 8   | CENTER     | 0.13 | Yes  | Structural / architectural grid lines and bubbles                 |
| A08  | ASG-REVISION          | 1   | Continuous | 0.25 | Yes  | Revision clouds and delta tags                                    |
| A09  | ASG-SYMBOL            | 3   | Continuous | 0.25 | Yes  | Section / detail / elevation marks, north arrow                   |
| S01  | ASG-BORDER            | 7   | Continuous | 0.50 | Yes  | Border frame in layouts                                           |
| S02  | ASG-TITLEBLOCK        | 7   | Continuous | 0.25 | Yes  | Title block insert and attributes                                 |
| S03  | ASG-VIEWPORT          | 6   | Continuous | 0.00 | NO   | All layout viewports                                              |
| S04  | ASG-NOLOT             | 6   | Continuous | 0.00 | NO   | Internal markups and reminders - never printed                    |

  

| ACI | Function                                                        | Plotted with ASG_Monochrome.ctb |
|-----|-----------------------------------------------------------------|---------------------------------|
| 7   | Primary geometry, metals, text, sheet                           | Black                           |
| 8   | Secondary geometry, hardware, seals, hidden, centrelines, grids | Black (lighter weights)         |
| 4   | Glass and mirror                                                | Black                           |
| 3   | Annotation: dimensions, leaders, levels, symbols                | Black                           |
| 1   | Revisions                                                       | Black                           |

| ACI | Function                                         | Plotted with ASG_Monochrome.ctb   |
|-----|--------------------------------------------------|-----------------------------------|
| 6   | Non-plot: construction, viewport, no-plot        | Not plotted (layer plot flag off) |
| 251 | Solid fills (aluminium / steel walls in section) | Grey RGB 91 (about 64 % dark)     |
| 252 | Hatch patterns, insulation                       | Grey RGB 132 (about 48 % dark)    |
| 253 | Reference background / xrefs                     | Grey RGB 173 (about 32 % dark)    |

## 6. Lineweight matrix (ISO 128 series) and linetypes

| Lineweight | Class       | Use                                                                            | Layers                                                                                                                                  |
|------------|-------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 0.50 mm    | Extra heavy | Section cut outlines, sheet border                                             | ASG-SECTION-CUT, ASG-BORDER                                                                                                             |
| 0.35 mm    | Heavy       | Primary visible outlines, cutting planes                                       | ASG-OUTLINE-PRIMARY, ASG-CUTTING-PLANE                                                                                                  |
| 0.25 mm    | Medium      | Aluminium, steel, cladding, symbols, revisions, title block                    | ASG-ALUMINIUM-PROFILE, ASG-STAINLESS-STEEL, ASG-MILD-STEEL, ASG-CLADDING, ASG-REVISION, ASG-SYMBOL, ASG-TITLEBLOCK                      |
| 0.18 mm    | Light       | Secondary lines, glass, hardware, fixings, hidden lines, text, tables          | ASG-OUTLINE-SECONDARY, ASG-GLASS, ASG-HARDWARE, ASG-FIXING, ASG-HIDDEN, ASG-TEXT, ASG-TEXT-NOTE, ASG-TABLE                              |
| 0.13 mm    | Fine        | Dimensions, leaders, centrelines, grids, seals, insulation, set-out, reference | ASG-GASKET, ASG-SEALANT, ASG-INSULATION, ASG-CENTERLINE, ASG-SETOUT, ASG-REFERENCE, ASG-DIMENSION, ASG-MULTILEADER, ASG-LEVEL, ASG-GRID |
| 0.09 mm    | Hatch       | Hatch patterns and fills only (also plotted grey)                              | ASG-HATCH, ASG-HATCH-FILL                                                                                                               |

Each class is about 1.4 x the next (ISO 128), so the hierarchy survives PDF and paper. 0.09 is used only for hatching, which also plots grey. LTSCALE 1 + PSLTSCALE 1 + MSLTSCALE 1: dash lengths are paper mm in every viewport and in model space.

| Linetype   | Pattern                   |
|------------|---------------------------|
| HIDDEN     | Hidden __ __ __           |
| HIDDEN2    | Hidden (.5x) _ _ _        |
| CENTER     | Center ____ _ ____        |
| CENTER2    | Center (.5x) ____ _ ____  |
| DASHED     | Dashed __ __              |
| DASHED2    | Dashed (.5x) _ _          |
| DASHDOT    | Dash dot __ . __          |
| DASHDOT2   | Dash dot (.5x) _ . _      |
| PHANTOM    | Phantom ____ _ ____       |
| PHANTOM2   | Phantom (.5x) ____ _ ____ |
| DOT        | Dot . . .                 |
| Continuous | Solid                     |

## 7. Text system - Arial (TrueType)

| Style            | Font        | Annotative       | Use                                                  |
|------------------|-------------|------------------|------------------------------------------------------|
| ASG-TEXT-ANNO    | arial.ttf   | Yes              | Model-space general text and dimension text          |
| ASG-TEXT-NOTE    | arial.ttf   | Yes              | Model-space notes, material labels, leaders          |
| ASG-TEXT-SMALL   | arial.ttf   | Yes              | Small detail notes, profile codes                    |
| ASG-TEXT-TITLE   | arialbd.ttf | No (paper space) | Detail / view titles (paper space)                   |
| ASG-TEXT-HEADING | arialbd.ttf | No (paper space) | Drawing titles, sheet headings (paper space)         |
| ASG-TEXT-SHEET   | arial.ttf   | No (paper space) | Title block fields, tables, sheet text (paper space) |

| Use                | Plotted height | Style                                 |
|--------------------|----------------|---------------------------------------|
| General dimensions | 2.5 mm         | ASG-TEXT-ANNO (via dimension style)   |
| Technical notes    | 2.5 mm         | ASG-TEXT-NOTE                         |
| Material labels    | 2.5 mm         | ASG-TEXT-NOTE (via ASG-ML-ANNO)       |
| Small detail notes | 1.8 mm         | ASG-TEXT-SMALL                        |
| Detail titles      | 3.5 mm         | ASG-TEXT-TITLE                        |
| Drawing titles     | 5.0 mm         | ASG-TEXT-HEADING                      |
| Sheet headings     | 7.0 mm         | ASG-TEXT-HEADING (A1); 5.0 on A3 / A4 |

All heights are from the ISO 3098 series; minimum 1.8 mm on any sheet. Arial is a Windows core font and embeds in PDF; if missing AutoCAD uses FONTALT (set arial.ttf).

## 8. Dimension system

| Style             | Annotative                                                                                                                                                        | Use                                                                                        |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| ASG-DIM-ANNO      | Yes                                                                                                                                                               | Default linear / aligned dimensions in model space, any viewport scale                     |
| ASG-DIM-FIXED     | No                                                                                                                                                                | Non-annotative: dimensions placed in paper space over viewports (true size), or 1:1 sheets |
| ASG-DIM-DETAIL    | Yes                                                                                                                                                               | Fabrication details 0.0; symmetrical tolerance switched on per dimension (height 0.7)      |
| ASG-DIM-ANGULAR   | Yes                                                                                                                                                               | Angular dimensions, decimal degrees 0.00                                                   |
| ASG-DIM-RADIUS    | Yes                                                                                                                                                               | Radius dimensions (R prefix automatic), no centre mark                                     |
| ASG-DIM-DIAMETER  | Yes                                                                                                                                                               | Diameter dimensions (diameter sign automatic), centre mark 2.5                             |
| Parameter         | Value                                                                                                                                                             |                                                                                            |
| Text              | ASG-TEXT-ANNO 2.5, above the dimension line (DIMTAD 1), aligned with it, gap 1.0                                                                                  |                                                                                            |
| Arrowheads        | Closed filled 2.5. Centre mark 2.5 (DIAMETER), none (RADIUS)                                                                                                      |                                                                                            |
| Extension lines   | Offset 1.5 from origin, extend 1.25 beyond dimension line                                                                                                         |                                                                                            |
| Spacing           | Baseline / continue spacing 7.0                                                                                                                                   |                                                                                            |
| Fit               | DIMATFIT 3 best fit; DIMTOFL 1 line drawn between extension lines; DIMTMOVE 1 leader added when text is moved (RADIUS / DIAMETER: dimension line moves with text) |                                                                                            |
| Linear units      | Decimal mm, '.' separator, precision 0 (DETAIL 0.0), rounding 0, trailing zeros suppressed (DETAIL keeps them: 50.0)                                              |                                                                                            |
| Angular units     | Decimal degrees, 0.00 on ASG-DIM-ANGULAR (trailing zeros shown)                                                                                                   |                                                                                            |
| Tolerance         | Off. On ASG-DIM-DETAIL switch on per dimension: symmetrical, precision 0.0, height 0.7                                                                            |                                                                                            |
| Alternate units   | Off                                                                                                                                                               |                                                                                            |
| Measurement scale | DIMLFAC 1.0 - always true geometry                                                                                                                                |                                                                                            |
| Scale             | Annotative styles take the annotation scale; ASG-DIM-FIXED DIMSCALE 1 for paper-space dims                                                                        |                                                                                            |
| Header            | Drawing's current dimension variables equal ASG-DIM-ANNO - no '<style overrides>' on new drawings                                                                 |                                                                                            |

Rule: dimensions always report measured geometry (DIMLFAC 1, DIMASSOC 2). Overriding dimension text is prohibited - fix the geometry. 'Standard' carries ASG-DIM-FIXED values so it is harmless if picked. No further fixed-scale styles are needed.

## 9. Multileader system

| Style            | Annotative | Arrow             | Text               | Frame | Use                                                  |
|------------------|------------|-------------------|--------------------|-------|------------------------------------------------------|
| ASG-ML-ANNO      | Yes        | closed filled 2.5 | ASG-TEXT-NOTE 2.5  | No    | Default model-space callouts and material labels     |
| ASG-ML-NOTE      | No         | closed filled 2.5 | ASG-TEXT-SHEET 2.5 | No    | Paper-space notes pointing into viewports            |
| ASG-ML-DETAIL    | Yes        | _DOT 1.5          | ASG-TEXT-SMALL 1.8 | No    | Dense 1:1 / 1:2 fabrication details                  |
| ASG-ML-REFERENCE | Yes        | _OPEN30 2.5       | ASG-TEXT-NOTE 2.5  | Yes   | References to other drawings / details (framed text) |

All: straight leader, maximum 2 points, landing gap 1.0, dogleg 4.0, text attached at the middle of the top line, ByLayer colour. Background mask is a property of each note (Properties > Text > Background mask, offset 1.2, use background colour) - use it where a note crosses hatch.

## 10. Annotative scale governance

Controlled list: 1:1, 1:2, 1:5, 1:10, 1:20, 1:25, 1:50, 1:100. Other scales only with CAD custodian approval (e.g. 1:200 site plans).

| Object           | Rule                                                                                         |
|------------------|----------------------------------------------------------------------------------------------|
| Model-space text | Annotative ASG-TEXT-ANNO / -NOTE / -SMALL - type the paper height (2.5 / 1.8)                |
| Paper-space text | Non-annotative ASG-TEXT-TITLE / -HEADING / -SHEET at true plotted height                     |
| Dimensions       | ASG-DIM-ANNO family in model space; ASG-DIM-FIXED for dimensions placed in paper space       |
| Multileaders     | ASG-ML-ANNO / -DETAIL / -REFERENCE in model space; ASG-ML-NOTE in paper space                |
| Symbols          | Section / detail / level / grid bubbles: annotative blocks                                   |
| Real-size blocks | Never annotative (hardware, profiles, fixings)                                               |
| Hatches          | Material hatches non-annotative (scale rule in section 13); symbolic fills may be annotative |

**Workflow:** 1) Set the viewport scale from the list, then lock the viewport. 2) Work inside the viewport - CANNOSCALE follows it. 3) Add annotation with ASG-\*-ANNO styles. 4) Same note needed at another scale: select it > right-click > Annotative Object Scale > Add (OBJECTSCALE). 5) ANNOAUTOSCALE 0 (profile) and ANNOALLVISIBLE 0 (drawing) prevent unwanted scales and visual duplicates; the status-bar 'Annotation Visibility' button shows all scales temporarily. 6) Drawings imported with extra scales: -SCALELISTEDIT > Reset, then AUDIT.

## 11. Tables and schedules

| Style              | Title      | Header     | Data   | Use                       |
|--------------------|------------|------------|--------|---------------------------|
| ASG-TABLE-STANDARD | 3.5 MC     | 2.5 MC     | 2.5 ML | Standard technical tables |
| ASG-TABLE-MATERIAL | 3.5 MC     | 2.0 MC     | 2.0 ML | Material schedules        |
| ASG-TABLE-GLASS    | 3.5 MC     | 2.0 MC     | 2.0 ML | Glass schedules           |
| ASG-TABLE-HARDWARE | 3.5 MC     | 2.0 MC     | 2.0 ML | Hardware schedules        |
| ASG-TABLE-REVISION | suppressed | 2.0 MC     | 2.0 ML | Revision registers        |
| ASG-TABLE-REGISTER | 3.5 MC     | 2.5 MC     | 2.5 ML | Drawing registers         |
| ASG-TABLE-NOTES    | 3.5 ML     | suppressed | 2.5 TL | General notes             |

Common: ASG-TEXT-SHEET, cell margins horizontal 1.5 / vertical 1.0, flow top-down, inside grid 0.18, outline 0.35, header fill ACI 254 (light grey), no data fill, colour ByBlock. Alignment codes: MC middle-centre, ML middle-left, TL top-left. Numbers centred, descriptions left. Tables go in paper space on ASG-TABLE. Created by the installer.

## 12. Layouts, page setups, title block and numbering

| Layout / named page setup          | Paper (canonical media name)           | Orient. | Title block | Default viewport |
|------------------------------------|----------------------------------------|---------|-------------|------------------|
| A4-LANDSCAPE /<br>ASG-A4-LANDSCAPE | ISO_full_bleed_A4_(297.00_x_210.00_MM) | L       | ASG-TB-A4-L | 1:10, locked     |
| A4-PORTRAIT /<br>ASG-A4-PORTRAIT   | ISO_full_bleed_A4_(210.00_x_297.00_MM) | P       | ASG-TB-A4-P | 1:10, locked     |
| A3-LANDSCAPE /<br>ASG-A3-LANDSCAPE | ISO_full_bleed_A3_(420.00_x_297.00_MM) | L       | ASG-TB-A3-L | 1:20, locked     |
| A3-PORTRAIT /<br>ASG-A3-PORTRAIT   | ISO_full_bleed_A3_(297.00_x_420.00_MM) | P       | ASG-TB-A3-P | 1:20, locked     |
| A1-LANDSCAPE /<br>ASG-A1-LANDSCAPE | ISO_full_bleed_A1_(841.00_x_594.00_MM) | L       | ASG-TB-A1-L | 1:50, locked     |

All page setups: DWG To PDF.pc3, plot area Layout, scale 1:1 (mm), offset 0,0, ASG\_Monochrome.ctb, plot object lineweights ON, lineweights not scaled, plot transparency OFF, paper space last, viewport borders not plotted. Border: 20 mm filing margin left, 10 mm other sides. Viewports on ASG-VIEWPORT, scale from the controlled list, Display Locked = Yes; unlock only to change the scale, then lock again.

| Title block | Type                          | Size mm     | Sections (top to bottom)                                                                                                                                                                                                         |
|-------------|-------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASG-TB-A4-L | Full-height strip (landscape) | 70 x 190.0  | Company - key plan / approval stamp - general notes + copyright - revisions (5 rows) - project, location, client, consultant - drawing title - drawn / checked / approved - status - scale / date - rev / sheet - drawing number |
| ASG-TB-A3-L | Full-height strip (landscape) | 98 x 277.0  | Company - key plan / approval stamp - general notes + copyright - revisions (5 rows) - project, location, client, consultant - drawing title - drawn / checked / approved - status - scale / date - rev / sheet - drawing number |
| ASG-TB-A1-L | Full-height strip (landscape) | 160 x 574.0 | Company - key plan / approval stamp - general notes + copyright - revisions (5 rows) - project, location, client, consultant - drawing title - drawn / checked / approved - status - scale / date - rev / sheet - drawing number |
| ASG-TB-A4-P | Bottom block (portrait)       | 180 x 106.5 | Company   approval stamp - notes + copyright   revisions (5 rows) - project   location - client   consultant - drawing title - drawn   checked   approved - scale   date   status - drawing number   rev   sheet                 |
| ASG-TB-A3-P | Bottom block (portrait)       | 267 x 120.2 | Company   approval stamp - notes + copyright   revisions (5 rows) - project   location - client   consultant - drawing title - drawn   checked   approved - scale   date   status - drawing number   rev   sheet                 |

Each title block is designed for its sheet - never stretched - and inserted at scale 1 at the border corner. Not explodable, uniform scale, mm units, attributes position-locked (edit with a double-click / EATTEDIT). Every caption, note and a design-length value for every field is measured with Arial metrics and fits its cell with at least 1 mm to spare. Company heading is text - no logo invented. 35 attributes: PROJECT, LOCATION, CLIENT, CONSULTANT, DWG\_TITLE, DWG\_TITLE\_2, DWG\_NO, REV, SHEET\_NO, SCALE, DATE, DWG\_STATUS, DRAWN\_BY, CHECKED\_BY, APPROVED\_BY, REV1\_NO, REV1\_DATE, REV1\_DESC, REV1\_BY, REV2\_NO, REV2\_DATE, REV2\_DESC, REV2\_BY, REV3\_NO, REV3\_DATE, REV3\_DESC, REV3\_BY, REV4\_NO, REV4\_DATE, REV4\_DESC, REV4\_BY, REV5\_NO, REV5\_DATE, REV5\_DESC, REV5\_BY. Drawing status values: PRELIMINARY, FOR APPROVAL, APPROVED, FOR CONSTRUCTION, FOR FABRICATION, AS-BUILT, SUPERSEDED.

| Drawing numbering |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| Format            | AS-[YYYY]-[NNN]-[DIS]-[TYP]-[SSS] + revision R0, R1, R2 ... in the REV field              |
| Example           | AS-2026-014-ALU-SD-001, REV R0 -> file AS-2026-014-ALU-SD-001-R0_MarinaTower.dwg          |
| YYYY-NNN          | AI Siraj project code (year + project serial)                                             |
| DIS               | ALU aluminium   GLZ glazing   MS mild steel   SS stainless steel   MISC other metal works |
| TYP               | GA general arrangement   EL elevation   SD shop drawing   SEC section   DT detail         |
| SSS               | Sheet serial 001, 002 ...                                                                 |

## 13. Hatch, block and attribute standards

| Material            | Pattern | Angle prop. | Model-space scale                     | Layer          | Note                                        |
|---------------------|---------|-------------|---------------------------------------|----------------|---------------------------------------------|
| Glass (section)     | ANSI31  | 0           | 0.5 x drawing scale (1.6 mm on paper) | ASG-HATCH      | Thin monolithic glass may be left unhatched |
| Aluminium (section) | SOLID   | -           | -                                     | ASG-HATCH-FILL | Plots dark grey                             |

| Material                      | Pattern | Angle prop. | Model-space scale                                              | Layer          | Note                                 |
|-------------------------------|---------|-------------|----------------------------------------------------------------|----------------|--------------------------------------|
| Mild steel (section)          | ANSI32  | 0           | 1.0 x drawing scale                                            | ASG-HATCH      | Lines read 45 deg                    |
| Stainless steel (section)     | ANSI31  | 90          | 0.75 x drawing scale                                           | ASG-HATCH      | Lines read 135 deg                   |
| Thermal break / polyamide     | ANSI37  | 0           | 0.5 x drawing scale                                            | ASG-INSULATION | Cross-hatch                          |
| Insulation board / batt       | INSUL   | 0           | insulation thickness / 9.525 (real size)                       | ASG-INSULATION | Pattern band fits the thickness      |
| Concrete                      | AR-CONC | 0           | real size: 1.0 up to 1:10, 2.0 at 1:20-1:25, 5.0 at 1:50-1:100 | ASG-REFERENCE  |                                      |
| Masonry / blockwork (section) | ANSI31  | 0           | 1.0 x drawing scale                                            | ASG-REFERENCE  | Plots light grey                     |
| Blockwork (elevation)         | AR-B816 | 0           | real size 0.984 (= 200 x 400 block)                            | ASG-REFERENCE  |                                      |
| Sealant bead                  | SOLID   | -           | -                                                              | ASG-SEALANT    | Plots black                          |
| Generic / unspecified section | ANSI31  | 0           | 1.0 x drawing scale                                            | ASG-HATCH      | Use only until material is confirmed |

Pattern modules (acadiso.pat, scale 1): ANSI31 / ANSI37 lines 3.175 mm apart, ANSI32 pairs 9.525 mm, INSUL band 9.525 mm, AR-\* patterns are real size. 'x drawing scale' patterns keep a constant spacing on paper. All pattern names verified to exist in acadiso.pat.

Blocks: name ASG-<CATEGORY>-<ITEM>-<VARIANT>, geometry on layer 0 ByLayer / ByBlock, base point at a datum (fixing point, bottom-left or centreline), block units mm, uniform scale, real-size items non-annotative, symbols annotative. Manufacturer data only from verified supplier sources.

| Attribute tag | Meaning                                      |
|---------------|----------------------------------------------|
| ITEM_CODE     | Item / product code                          |
| ITEM_NAME     | Item description                             |
| SYSTEM_NAME   | System / series                              |
| MANUFACTURER  | Manufacturer / supplier (verified data only) |
| MATERIAL      | Base material                                |
| FINISH        | Finish                                       |
| COLOUR        | Colour / RAL                                 |
| GLASS_SPEC    | Glass specification                          |
| REVISION      | Block revision                               |

## 14. Plotting / PDF standard

| Item                   | Standard                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------|
| Device                 | DWG To PDF.pc3 (ships with AutoCAD) - no custom PC3 dependency                            |
| Paper                  | ISO full bleed A4 / A3 / A1 media = exact sheet size; the border provides the margins     |
| Area / scale / offset  | Layout, 1:1 (mm), offset 0,0 ('Center the plot' is not available for Layout)              |
| Plot style             | ASG_Monochrome.ctb - black; ACI 250-254 keep their grey. Fallback: monochrome.ctb         |
| Lineweights            | Plot object lineweights ON, scale lineweights OFF, plot paper space last                  |
| Transparency / shading | Plot transparency OFF; shaded viewports 'As displayed', presentation quality              |
| PDF options            | Vector 1200 dpi, raster 300 dpi, TrueType as text, layer information OFF for client issue |
| Pre-issue              | AUDIT, PURGE, plot preview, open the PDF and check scale, dimensions and no-plot items    |
| Compatibility          | AutoCAD 2018 DWG (opens in AutoCAD 2018-2026); DXF 2018 for third parties; no proxies     |
| Xrefs / PDF import     | Relative paths, overlay, layer ASG-REFERENCE; PDFIMPORT geometry moved to ASG layers      |

## 15. Resource separation

| Resource                                                                         | Where it lives                       | Deploy                                              |
|----------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------|
| Layers, styles, scales, layouts, title blocks, header variables                  | Inside the DWT                       | Copy DWT to shared Templates folder                 |
| Table styles, named page setups, CANNOSCALE / MSLTSCALE / HPLAYER / DIMLAYER ... | Inside the DWT after the installer   | Travel with the DWT                                 |
| ASG_Monochrome.ctb                                                               | Plot Styles folder (external)        | Copy to each PC or a shared support path            |
| DWG To PDF.pc3                                                                   | Plotters folder (ships with AutoCAD) | None                                                |
| arial.ttf / arialbd.ttf                                                          | Windows Fonts (external)             | Present on Windows; FONTALT = arial.ttf as fallback |
| Linetypes                                                                        | Embedded in the DWT (acadiso values) | None                                                |
| Hatch patterns                                                                   | acadiso.pat (ships with AutoCAD)     | None                                                |
| OSMODE, grips, autosave, pickbox ...                                             | User profile / registry              | Optional ASG-PROFILE per user                       |
| QNEW default template                                                            | Options > Files (profile)            | Set per PC to ASG_Master_Template.dwt               |

## 16. Portability checklist (each PC)

- [ ] AutoCAD 2018 or later installed; DWG To PDF.pc3 present (default).
- [ ] 09\_Team\_Kit copied to the PC or reachable on the shared drive.
- [ ] ASG\_Install.lsp dragged into AutoCAD, ASG\_Master\_Template.dwt selected: popup reads ASG INSTALL COMPLETE.
- [ ] Ctrl+N opens a drawing with ASG- layers and the five sheet tabs.
- [ ] Arial installed (Windows default); FONTALT = arial.ttf.
- [ ] Optional: user runs ASG-PROFILE once (snaps, grips, autosave).

## 17. Installing and verifying in AutoCAD

1. Extract the package zip. Double-click 01\_Master\_Template\ASG\_Master\_Template.dxf to open it in AutoCAD.
2. Drag 01\_Master\_Template\ASG\_Install.lsp into the AutoCAD window; click 'Load Once' if AutoCAD asks.

3. The installer copies the CTB, creates table styles / page setups / settings, runs ASG-QA, saves the DWT into AutoCAD's Template folder and the DWG master copy, fills 09\_Team\_Kit and sets Ctrl+N.
4. Read the popup: ASG INSTALL COMPLETE, or the exact items to do by hand. Log: ASG\_QA\_Log.txt.
5. Open QA\_TEST\_DRAWING\_temporary.dxf, plot A3-LANDSCAPE to PDF: check dims, no 'NOPLOT' text / diagonal, no viewport frames, 6 lineweights distinguishable, text legible.
6. Ctrl+N: annotative text / dim / leader at 1:20, then add 1:5 via OBJECTSCALE - confirm 2.5 mm on paper. Do not save the test.

Appendix - layout previews (rendered from the saved master DXF)

A4-LANDSCAPE

|                                                                      |            |             |    |
|----------------------------------------------------------------------|------------|-------------|----|
| SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.                        |            |             |    |
| AL SIRAJ GROUP - ALUMINIUM, GLASS & FACADE DIVISION - UAE            |            |             |    |
| APPROVAL STAMP                                                       |            |             |    |
| GENERAL NOTES                                                        |            |             |    |
| 1. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS NOTED OTHERWISE.         |            |             |    |
| 2. DO NOT SCALE THIS DRAWING - USE FIGURED DIMENSIONS ONLY.          |            |             |    |
| 3. VERIFY ALL DIMENSIONS ON SITE BEFORE FABRICATION.                 |            |             |    |
| 4. READ WITH ARCHITECTURAL, STRUCTURAL DRAWINGS AND SPECIFICATIONS.  |            |             |    |
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| REV                                                                  | DATE       | DESCRIPTION | BY |
| R0                                                                   | DD.MM.YYYY | FIRST ISSUE | -  |
|                                                                      |            |             |    |
|                                                                      |            |             |    |
|                                                                      |            |             |    |
|                                                                      |            |             |    |
| PROJECT                                                              |            |             |    |
| PROJECT NAME                                                         |            |             |    |
| LOCATION                                                             |            |             |    |
| LOCATION, UAE                                                        |            |             |    |
| CLIENT                                                               |            |             |    |
| CLIENT                                                               |            |             |    |
| CONSULTANT                                                           |            |             |    |
| CONSULTANT                                                           |            |             |    |
| DRAWING TITLE                                                        |            |             |    |
| DRAWING TITLE                                                        |            |             |    |
| DRAWN BY                                                             |            | CHECKED BY  |    |
| -                                                                    |            | -           |    |
| APPROVED BY                                                          |            |             |    |
| -                                                                    |            |             |    |
| DRAWING STATUS                                                       |            |             |    |
| PRELIMINARY                                                          |            |             |    |
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| R0                                                                   | 01 OF 01   |             |    |
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A4-PORTRAIT

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|              |               |
|--------------|---------------|
| PROJECT      | LOCATION      |
| PROJECT NAME | LOCATION, UAE |
| CLIENT       | CONSULTANT    |
| CLIENT       | CONSULTANT    |

DRAWING TITLE

DRAWING TITLE

|          |            |                |
|----------|------------|----------------|
| DRAWN BY | CHECKED BY | APPROVED BY    |
| -        | -          | -              |
| SCALE    | DATE       | DRAWING STATUS |
| AS SHOWN | DD.MM.YYYY | PRELIMINARY    |

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| DRAWING No.             | REV | SHEET No. |
| AS-YYYY-NNN-DIS-TYP-001 | R0  | 01 OF 01  |

A3-LANDSCAPE

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KEY PLAN

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PROJECT

PROJECT NAME

LOCATION

LOCATION, UAE

CLIENT

CLIENT

CONSULTANT

CONSULTANT

DRAWING TITLE

DRAWING TITLE

DRAWN BY

CHECKED BY

APPROVED BY

DRAWING STATUS

PRELIMINARY

SCALE

DATE

AS SHOWN

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REV

SHEET No.

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A3-PORTRAIT

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|-------------------------|---------------|----------------|
| PROJECT                 | LOCATION      |                |
| PROJECT NAME            | LOCATION, UAE |                |
| CLIENT                  | CONSULTANT    |                |
| CLIENT                  | CONSULTANT    |                |
| DRAWING TITLE           |               |                |
| DRAWING TITLE           |               |                |
| DRAWN BY                | CHECKED BY    | APPROVED BY    |
| SCALE                   | DATE          | DRAWING STATUS |
| AS SHOWN                | DD.MM.YYYY    | PRELIMINARY    |
| SUBJECT                 |               | SHEET NO.      |
| AS-YYYY-NNN-DIS-TYP-001 |               | 01 OF 01       |
|                         |               | R0             |

A1-LANDSCAPE

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KEY PLAN

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PROJECT

PROJECT NAME

LOCATION

LOCATION, UAE

CLIENT

CLIENT

CONSULTANT

CONSULTANT

DRAWING TITLE

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DRAWING STATUS

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